

Hi-Singers: A Comprehensive High-Quality Dataset for Expressive Audio-Driven Singing Head Synthesis

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Abstract

State-of-the-art models for audio-driven digital human generation have achieved photo-realistic results in talking-head synthesis. However, extending these models to singing-head synthesis remains challenging due to a significant Domain Gap: singing requires more exaggerated expressions, vivid jaw openings, and precise rhythmic synchronization. Current models, primarily trained on speech datasets, often struggle with “rhythmic drift” and constrained dynamics. To address this, we introduce Hi-Singers, the first large-scale, high-quality, in-the-wild video dataset specifically tailored for singing head synthesis. Hi-Singers undergoes a rigorous automated and manual filtering pipeline, ensuring strict thematic adherence, high-resolution rendering, and stable motion, resulting in 29,608 video segments totaling approximately 170 hours. We further establish a dedicated evaluation benchmark balanced across linguistic and musical styles. Extensive experiments across diverse architectures—including 3D-coefficient and diffusion-based models—demonstrate that Hi-Singers consistently and significantly improves performance across all dimensions. Specifically, it enables models to achieve superior visual realism, enhanced lip-sync consistency, and more precise rhythmic dynamics, effectively bridging the domain gap and setting a new performance standard for the singing synthesis task. The dataset is available at <https://huggingface.co/datasets/CharlesZhang-USTC/Hi-Singers>.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

Video Datasets, Singing Head Synthesis, Expressive Facial Dynamics

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1 Introduction

Audio-driven digital human generation aims to synthesize photo-realistic facial animations based on audio input and a reference character portrait. While significant progress has been made in “Talking Head” synthesis by prioritizing lip-sync and audio consistency, the extension to “Singing Head” synthesis presents a distinct set of challenges that remain largely unresolved.

The primary hurdle lies in the significant **Domain Gap** between speech and singing. First, singing audio is characterized by complex melodies and instrumental backgrounds, which often interfere with accurate audio-visual alignment. Second, singing requires far more exaggerated facial expressions and vivid jaw openings compared to the relatively static nature of speech. Finally, singing is inherently rhythmic; current talking-head models, when applied to singing, often suffer from “rhythmic drift” and lack the precise music-aligned dynamics necessary for realism.

Existing state-of-the-art models, such as EMO [20] and Hallo [24], typically employ identical architectures for both tasks, relying solely on data to bridge the gap. However, current datasets are heavily biased toward talking content. For instance, HDTF [26] focuses on front-facing talk shows, while Talkvid [3] emphasizes linguistic diversity in speech. Specialized resources for singing are remarkably scarce. Existing singing datasets often suffer from small quantities, lack photorealistic 2D textures (focusing instead on 3D mesh), or are restricted to controlled laboratory settings that fail to generalize to in-the-wild scenarios.

Regarding the issues raised above, the main contributions of this paper are summarized as follows:

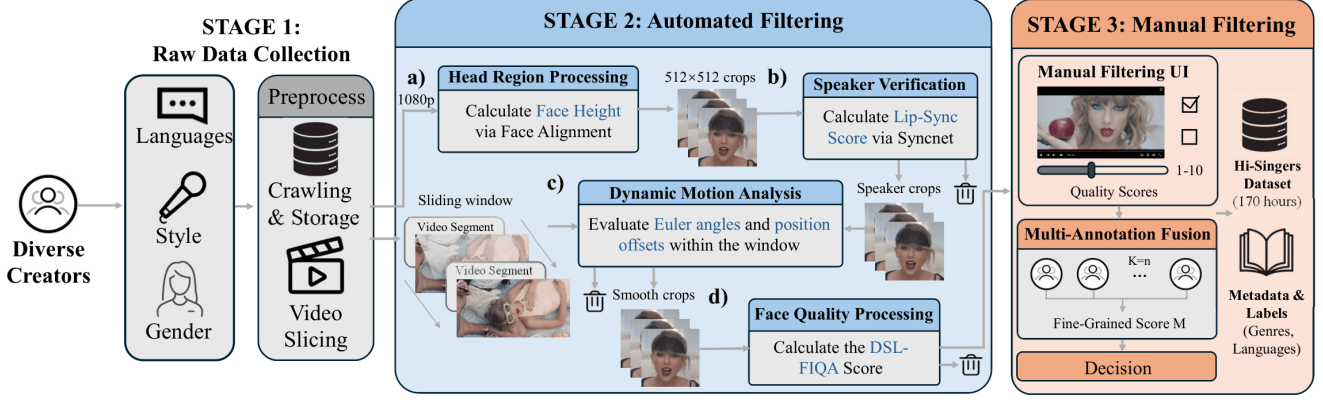


Figure 1: The framework diagram of the data processing workflow illustrates the entire process from raw internet data to the final dataset. This process can be divided into the following three stages: 1) Data collection and preprocessing 2) Automated filtering 3) Manual filtering.

- **Hi-Singers Dataset:** We introduce the first large-scale, high-quality, in-the-wild video dataset specifically tailored for 2D singing head synthesis. It comprises 29,608 video segments totaling approximately 170 hours, filtered through a rigorous pipeline to ensure stable motion and high-quality rendering.
- **Domain-Specific Benchmark:** We construct a dedicated evaluation benchmark balanced across diverse linguistic (English and Chinese) and musical styles (Pop, Rock, Classical, etc.), providing a standardized platform for the community.
- **Rhythmic Evaluation Metric:** To quantitatively measure singing-specific performance, we introduce the **Beat Alignment Score (BAS)**. Unlike traditional lip-sync metrics, BAS quantifies the rhythmic synchronization between audio beats and kinematic motion peaks, effectively identifying the “rhythmic drift” common in generic models.
- **Experimental Validation:** Comprehensive experiments across diverse SOTA architectures—including 3D-coefficient-based (SadTalker) and Latent Diffusion (V-Express)—demonstrate that training on Hi-Singers significantly improves lip-sync consistency and captures precise, music-aligned rhythmic dynamics.

2 Related Works

Audio-Visual Synchronized Human Animation Synthesis. Synthesis models have evolved from complex 3D-based pipelines to 2D end-to-end approaches [27]. Early 3DMM-based methods mostly decouple intermediate representations of facial information (e.g., VOCA [7], FLAME [14]). Recent NeRF-based [9, 12] and 3D Gaussian Splatting (3DGS) methods [11, 18] have achieved high precision in synchronization.

2D generative models have transitioned from GAN-based approaches (e.g., Wav2Lip [19]) to diffusion-based architectures. State-of-the-art models such as EMO [20], Hallo [24], and V-Express [21] directly map audio features to realistic sequences. However, as noted in recent studies, these generic models are primarily trained

on talking-head data and often struggle with the exaggerated expressions and rhythmic synchronization inherent in singing.

Datasets for Human Animation. The evolution of datasets reflects a transition from lab-based recordings (e.g., GRID [6]) to large-scale in-the-wild samples (e.g., VoxCeleb [17], VFHQ [22]). For high-resolution tasks, HDTF [26] and TalkVid [3] provide diverse speech samples.

Table 1: Comparison of existing singing-related datasets. Most prior works focus on 3D animation or small-scale lab environments, leaving a gap for large-scale 2D video generation.

Dataset	Modality / Task	Setting	Res.	Scale
RAVDESS [16]	Emotion / Speech	Lab	720p	~1 hr
MusicFace [15]	3D Mesh / Anim.	Studio	-	~40 hrs
Song2Face [10]	3D Animation	Lab	-	~2 hrs
SingingHead [8]	4D Scan / 3D Anim.	Lab	1024p	~27 hrs
ChorusHead [23]	3D Head Anim.	In-the-wild	-	~8 hrs
RapVerse [2]	Body Motion / 3D	In-the-wild	-	~27 hrs
Ours	2D Video Gen.	In-the-wild	1080p	170 hrs

Despite these advancements, existing singing-specific datasets often focus on 3D parameters or body motion (e.g., MusicFace, RapVerse), lacking the photorealistic textures required for 2D video generation. Furthermore, lab-collected data like RAVDESS lacks the environmental diversity critical for real-world generalization. This leaves a significant resource gap for high-quality, large-scale, in-the-wild 2D singing head data.

3 Principles for Dataset Construction

As discussed in Section 1, datasets designed specifically for generating singing head tasks present greater challenges. For this reason, we have established rigorous principles for dataset construction.

Centered on Singing. The collected videos must be facial recordings demonstrating human singing actions. Irrelevant segments—including non-melodic talking segments, segments with background music only that show weak correlation with facial movements, and facial actions occurring between singing segments—must be excluded.

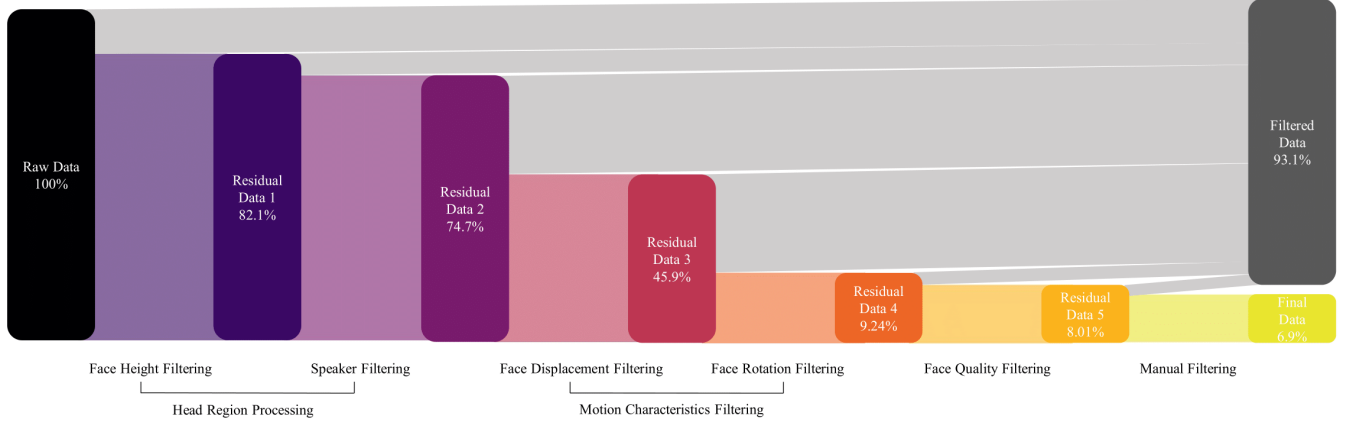


Figure 2: This Sankey diagram illustrates the flow of data during the processing stages. Specifically, it shows the data filtered out and retained at each step of the filtering process.

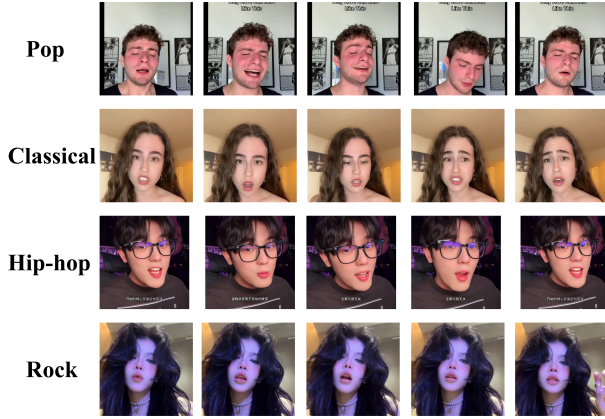


Figure 3: This diagram displays sample video frames representing the primary music genres within the dataset (pop, classical, hip-hop and rock).

Sufficient Facial Resolution. The resolution of faces detected in publicly available videos often varies significantly, and the resolution of the original collected videos may also differ due to platform limitations. The resolution of the face region in the filtered video should be sufficiently high and stable.

Complete Display of Facial Features. A complete display of facial features means that all facial elements should be fully visible without distortion or obstruction. During dataset construction, faces obscured by props such as masks, profiles that do not show complete facial features, or partially blurred faces must be filtered out.

Stability and Continuity of Facial Display. Singing videos often feature accompanying actions such as dancing or playing musical instruments, which can cause significant facial displacement and rotation. When constructing the dataset, it is essential to ensure that the face remains within the camera frame at all times and is free from excessive shaking or displacement.

4 Dataset Construction Pipeline

An intuitive framework diagram of the dataset processing workflow is shown in Figure 1, comprising three stages: data collection and preprocessing, automated filtering, and manual filtering.

4.1 Raw Data Collection

Approximately 1,200 gigabytes of raw video content totalling 4,500 hours were collected from video platforms including YouTube, TikTok (Douyin) and Bilibili. During the collection process, adherence to rigorous selection criteria was maintained, encompassing the gathering of front-facing singing videos of online singing influencers, with the objective of ensuring the isolation of high-quality singing faces. The present focus of this study is on videos in both English and Chinese languages, encompassing a variety of ethnicities, singing genres (e.g. pop, classical, rock, hip-hop) and accompaniment types (e.g. a cappella, original backing tracks, remixes). The identification of distinct singing languages and musical styles is achieved through the analysis of vocal bloggers from various geographical regions. Thereafter, the raw data underwent a series of preprocessing steps. Initially, the selection of videos was restricted to those with a resolution of 1080p or higher. Secondly, the video frame rate was adjusted to 25 fps, and the audio sampling rate to 16kHz. Finally, the encoding format was converted to H.264, and the video was sliced into segments for easier subsequent processing.

Following the collection and preprocessing of the raw data, both automated and manual filtering processes are conducted. The evaluation metrics for filtering are categorised into two approaches: **Threshold-Based Filtering (TBF)** and **Scoring-Based Filtering (SBF)**. The employment of TBF metrics facilitates the elimination of factors that objectively impact quality, including resolution, head displacement, and head shaking. Conversely, SBF metrics are utilised to evaluate the aesthetic and artistic quality of videos, encompassing factors such as facial clarity, vocal proficiency, and articulation clarity. Figure 2 illustrates the data flow at each step of the dataset processing pipeline.

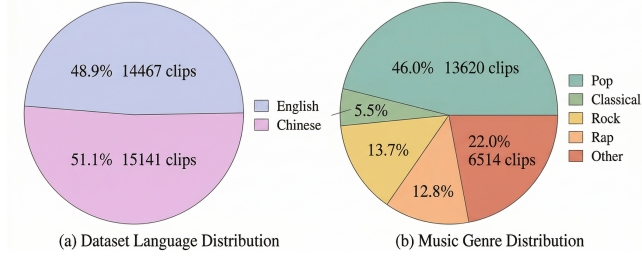


Figure 4: This chart presents statistical data on the distribution of languages and music genres within the dataset. It is evident that the dataset contains roughly equal numbers of English and Chinese language entries; the music genres are more diverse, with pop music dominating as the predominant style.

4.2 Automated Processing Pipeline

This section introduces the industrial-grade automated video processing and screening pipeline, which encompasses the following processes.

Stage1: Head Region Processing. To more conveniently identify faces within the dataset and their complex positional information, we employed the accurate face alignment network face-alignment [1]. The 2D version of face-alignment is employed to determine face height, filtering out faces with insufficient height (TBF) at a sufficient confidence level. Such faces are deemed to have inadequate clarity. Next, we perform a crop operation on the video based on the average center position of the face. The processed video resolution is 512×512 , containing only one face. To ensure the one-to-one correspondence between the face and the audio, we use SyncNet [5] to detect lip-sync alignment (TBF), eliminating faces that do not belong to the video’s audio.

Stage2: Motion Characteristics Filtering. The objective of this stage is to achieve the stable, forward-facing singing head shot. A sliding window approach is employed to analyse the displacement of facial centres within video footage. Samples are discarded if the mean displacement (TBF) across all windows in the entire video is excessively large, which excludes videos featuring significant facial movement. Next, we define the faces in the video frame that are completely vertical facing the camera as the “standard front face.” The 3D version of face-alignment is utilised for the extraction of facial landmarks, with the solvePnP algorithm then being applied to compute the yaw, pitch, and roll of Euler angles of detected faces relative to a “standard front face.” Video clips with excessive angles (TBF) will be filtered out, as they are deemed to lack stable and sufficient facial information. Figure 5 details the meaning of facial Euler angles and provides examples of videos that fail to meet the filtering criteria.

Stage3: Face Quality Processing. In this phase, DSL-FIQA [4] was employed for the purpose of facial image quality assessment. Specifically, the video frames were sampled at a rate of 5 frames per second, thereby generating a sequence of scores. Subsequently, the mean and interquartile range (IQR) of the sequence scores were calculated. The two scores are used both TBF to filter out videos that



Figure 5: The reference coordinate system for facial Euler angles (yaw, pitch, and roll) is illustrated in the leftmost diagram. The three scenarios requiring filtering are depicted in the subsequent diagrams, each representing excessive angular deviations relative to a face perfectly aligned with the camera within the coordinate system.

Table 2: Tables for the filtering stage, metrics, calculation, and filtering approach (TBF or SBF)

Category / Metrics	Calculation	Approach
1. Head Region Processing		
Face Height	Key Points	TBF
Lip-sync Alignment	SyncNet Score	TBF
2. Motion Characteristics Filtering		
Facial Movement	Mean Displacement	TBF
Facial Rotation	Face Euler Angle	TBF
3. Face Quality Processing		
Face Quality	DSL-FIQA Score	TBF/SBF
4. Manual Filtering		
Singing Content Relevance	Manual	SBF
Aesthetic Quality	Manual	SBF
Movement Stability	Manual	SBF
Articulation Clarity	Manual	SBF

fail to meet requirements and SBF to evaluate the overall quality of videos.

Following the completion of processing pipelines across the aforementioned three stages, all video footage has been segmented into clips ranging from 20 to 25 seconds in duration. The number of processed video clips is 29,608, with a total duration of approximately 170 hours. The statistical distribution of languages and genres within the dataset is illustrated in Figure 4.

4.3 Manual Filtering and Scoring Strategy

To ensure high-quality data and human-centric evaluation, we implement a multi-stage manual screening process. This process serves two primary functions: filtering out videos where the automated processing pipeline fails and providing a subjective quality assessment from a human perspective.

Five Scoring-Based Filtering (SBF) metrics were selected for their strong correlation with human perception: singing content relevance (SR), aesthetic quality (AQ), movement stability (MS), articulation clarity (AC), and the DSL-FIQA score (DF). A panel of ten experienced engineers scored a representative subset to establish a gold standard for parameter estimation.

We employ a logistic regression model to learn the optimal weights w^* for these metrics by mapping average scores $\bar{x}_{mn}$ to overall human acceptance decisions Y_m . The final comprehensive video quality score S_m for each video is defined as:

$$S_m = \sum_{n=1}^N w_n^* \cdot \bar{x}_{mn} \quad (1)$$

This weighted scoring scheme was subsequently applied to the entire dataset to determine the final inclusion of video samples. This strategy ensures that the resulting Hi-Singers dataset adheres to high expressive and aesthetic standards.

5 Experiments

5.1 Experimental Settings

5.1.1 Model and Baselines. To validate the effectiveness and generalizability of Hi-Singers across different generative paradigms, we conduct benchmarks using three state-of-the-art (SOTA) architectures:

- **Hallo** [24]: A hierarchical diffusion-based model capable of handling both speaking and singing face generation.
- **SadTalker** [25]: A 3D-coefficient-based architecture used to evaluate the impact of Hi-Singers on structural facial modeling.
- **V-Express** [21]: A latent diffusion-based model representing the latest advancements in audio-to-video mapping.

We specifically compare models trained under four distinct conditions to evaluate the domain gap: (1) **Hallo (Original)** trained on standard talking and singing data; (2) **Hallo (CelebV-HQ)** trained on talking-face centric data [28]; (3) **Additional SOTA Baselines** using official pre-trained weights from SadTalker and V-Express; and (4) **Hallo (Hi-Singers)** trained on our curated subset designed for vocal facial analysis.

Due to the lack of existing singing-specific benchmarks, we constructed a test set from the reserved portion of Hi-Singers, comprising 200 clips balanced across languages (Chinese, English) and genres (Pop, Classical, Rock, Hip-hop, etc.).

5.1.2 Implementation Details. To ensure a fair comparison, all experiments follow the original training protocols and hyperparameter settings. Training was executed on a cluster equipped with 8 NVIDIA A100 GPUs. Both training stages utilized a fixed learning rate of $1e^{-5}$ with the AdamW optimizer. The motion module was

initialized with pre-trained weights from AnimateDiff. The complete procedure required approximately one week for convergence per model configuration.

5.1.3 Evaluation Metrics. We assess performance across multiple dimensions:

- **Visual Quality:** Fréchet Inception Distance (**FID**) evaluates individual frame realism, while Fréchet Video Distance (**FVD**) assesses temporal consistency.
- **Audiovisual Synchronization:** We employ **Sync-C** and **Sync-D** to quantify lip-sync accuracy.
- **Rhythmic Dynamics:** We introduce the **Beat Alignment Score (BAS)** [13] to quantify synchronization between facial motion and music beats. BAS measures the Gaussian-weighted distance between audio beats and kinematic velocity peaks with a strict constraint ($\sigma = 0.2$), penalizing motion offsets exceeding ~ 200 ms.
- **Subjective Evaluation:** A 5-point **Mean Opinion Score (MOS)** is assigned by 10 independent researchers to assess naturalness and overall performance.

5.2 Quantitative Results and Analysis

Overall Performance. As demonstrated in Table 3, the model trained on Hi-Singers achieves SOTA performance across all dimensions. Notably, it not only leads in synchronization (7.72 Sync-C) and rhythmic precision (**0.168 BAS**) but also significantly improves visual quality, achieving the lowest **FID (23.15)** and **FVD (181.42)**. This comprehensive superiority confirms that Hi-Singers provides essential motion priors and high-fidelity textures that generic talking-head datasets lack.

Table 3: Overall performance comparison on singing head generation. Hi-Singers achieves SOTA results across all dimensions. ‘Orig.’ denotes official weights trained on generic talking datasets.

Method	FID↓	FVD↓	Sync-C↑	BAS↑	MOS↑
<i>Additional SOTA Baselines (Domain Gap Evaluation)</i>					
SadTalker [25] (Orig.)	38.54	245.12	6.85	0.052	3.12
V-Express [21] (Orig.)	25.68	192.45	6.92	0.076	3.45
<i>Main Results and Ablations</i>					
Hallo (Original Weights)	23.21	188.34	7.58	0.139	3.85
Hallo (CelebV-HQ [22])	42.23	385.67	7.12	0.111	3.63
Hallo (Hi-Singers)	23.15	181.42	7.72	0.168	3.92

Generalization Capabilities. Table 4 illustrates the robust performance of Hi-Singers across different linguistic and musical domains. Models trained on our dataset maintain high lip-sync consistency (Sync-C) and rhythmic alignment (BAS) in both Chinese and English, as well as across dynamic genres like Pop and Rock. These results validate the effectiveness of our specialized filtering pipeline in capturing the complex, high-variance motion patterns inherent in singing performance.

Table 4: Detailed Cross-Domain Analysis. We analyze the performance of **Hallo (Original)** and **Hallo (Hi-Singers)** across different categories. The averages of sub-categories are fitted to the overall performance results. **BAS** quantifies rhythmic synchronization.

Training Dataset	Metric	Language		Music Style				
		Chinese	English	Pop	Classical	Rock	Hip-hop	Other
Hallo (Original)	FID ↓	23.30	23.12	23.10	23.25	23.18	23.30	23.22
	FVD ↓	190.18	186.50	187.20	189.50	187.80	188.50	188.70
	Sync-C ↑	7.50	7.66	7.54	7.55	7.40	7.85	7.56
	Sync-D ↓	7.98	7.58	7.90	7.85	8.02	7.25	7.88
	BAS ↑	0.136	0.142	0.140	0.135	0.138	0.145	0.137
	MOS ↑	3.82	3.88	3.80	3.85	3.75	3.88	3.97
Hallo (Hi-Singers)	FID ↓	23.10	23.20	23.05	23.15	23.12	23.22	23.21
	FVD ↓	175.64	187.20	178.50	181.20	182.10	183.00	182.30
	Sync-C ↑	7.82	7.62	7.75	7.72	7.73	7.66	7.74
	Sync-D ↓	7.78	7.56	7.68	7.62	7.66	7.65	7.44
	BAS ↑	0.170	0.166	0.170	0.165	0.172	0.166	0.167
	MOS ↑	3.94	3.90	3.95	3.92	3.88	3.90	3.95

5.3 Qualitative Comparison

Figure 6 highlights the visual gap between models trained on generic talking datasets versus our proposed Hi-Singers. The **Baseline** (Top) suffers from rigid lips and blur, particularly during high-pitched notes where mouth movements are most extreme. In contrast, **Ours** (Bottom) produces **vivid large jaw openings** and sharp rhythmic motion. This visual evidence validates that Hi-Singers provides critical motion priors for expressive singing—such as intense articulation and rhythmic head bobs—that are absent in standard talking-head datasets.

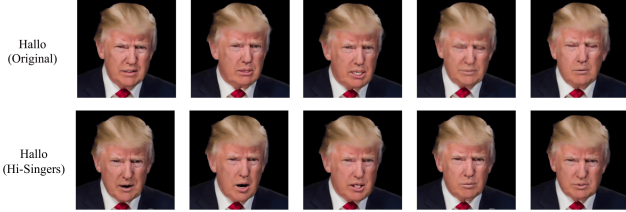


Figure 6: Qualitative Comparison. *Top:* Baseline trained on generic datasets shows constrained motion and blur. *Bottom:* Ours trained on Hi-Singers displays vivid jaw openings and sharpness.

5.4 Analysis of Experimental Results

5.4.1 Overall Performance and Domain Gap. As shown in Table 3, **Hallo (Hi-Singers)** achieves SOTA results across all dimensions, significantly leading in synchronization (7.72 Sync-C) and rhythmic precision (0.168 BAS). It also maintains superior visual fidelity with the lowest FID (23.15) and FVD (181.42), proving that high-expressivity singing does not compromise frame-level realism. In contrast, the poor performance of generic baselines (**SadTalker** and **V-Express**) in BAS (0.052 and 0.076) reveals a severe domain gap, where their rigid or misaligned motions justify the necessity of Hi-Singers for rhythmic-aware singing synthesis.

5.4.2 Cross-Linguistic Generalization. The analysis in Table 4 reveals distinct patterns across linguistic boundaries. Hi-Singers achieves a comprehensive lead in Chinese singing synthesis, reaching top

scores in both **Sync-C** (7.83) and **BAS** (0.172), which demonstrates its efficacy in capturing complex tonal-linguistic articulations. While **Hallo (Original)** shows a slight advantage in English temporal consistency (186.50 FVD) due to the vast distribution of English speech in generic datasets, Hi-Singers maintains superior lip-sync and rhythmic precision, offering a more balanced performance across multi-lingual content.

5.4.3 Adaptability to Musical Styles. The analysis across genres in Table 4 highlights how data diversity impacts synthesis quality. Hi-Singers dominates high-variance categories such as Pop and Rock, reaching the highest rhythmic scores (0.171 Rock BAS) and subjective quality (3.95 Pop MOS). Interestingly, while **Hallo (Original)** excels in **Hip-hop** lip-sync (7.85 Sync-C) due to its inherent similarity to fast-paced speech, Hi-Singers still leads in **BAS**, ensuring that motions remain strictly beat-aligned. For the **Other** category, Hi-Singers demonstrates broad adaptability across unconventional musical structures, showcasing its robust generalization capabilities.

6 Conclusion

To address the data bottleneck in singing head synthesis, we introduce **Hi-Singers**, a large-scale, high-quality dataset with 170 hours of rigorously filtered, in-the-wild footage, filling a critical gap left by generic talking-head datasets. Beyond data collection, we contribute an evaluation benchmark and the **Beat Alignment Score (BAS)**, a specialized metric for rhythmic synchronization. Experimental results across state-of-the-art architectures demonstrate that Hi-Singers enables superior performance in all dimensions, particularly visual realism, lip-sync accuracy, and rhythmic precision. The Hi-Singers framework provides a robust foundation for expressive digital human synthesis, steering the field toward synchronized and natural vocal performances.

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